Introduction to Software Engineering

COSC 336

ReSource (Phases IV and V)

Design Document

Architecture Design and Detailed Design

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Dr. Ala' Alsamarneh
Dr. Hamdan Alshkeili
Dr. Natnael Takele
Date: 03/11/2026

Prepared by: Group 3
Rashid Alzaabi 100064869

Prepared for:
Head of Facilities Management

Proposed Technical Owner:
University IT Department

Sections in This File

This file contains the parts of the ReSource Phase 4 and 5 Design Document prepared by Rashid Alzaabi (100064869). Section, figure and table numbers are the same as in the combined group document.

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- 7.7 AI Report Review
- 9.4 Notifications, Audit and Sustainability
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6. Security

Prepared by: Rashid Alzaabi

6.1 Security Boundaries

Prepared by: Rashid Alzaabi

Table 6-1 lists the five places where data crosses from one level of trust to another, and the control at each. The numbers match the security column (7) and the integration column (5) in Figure 3-1.

Table 6-1: Security boundaries

Boundary	Crossing	Controls	Requirements
B1 Internet to server	Browser to the reverse proxy	HTTPS only (TLS 1.2 or later), HTTP redirected; secure, HTTP-only session cookie; 30-minute idle limit; lockout after 5 failed sign-ins	SEC-1, SEC-3, SEC-4, SEC-6
B2 User to function	Every service call inside the application	Default deny; the role, department and account status are checked on each call (Table 6-2)	FR-6.7, SEC-5
B3 Application to database	Web and worker containers to MySQL	Private network only; separate database accounts; the audit table can only be inserted into	FR-7.A2, SEC-12
B4 ReSource to AI Provider	AIGateway to the AI Provider	HTTPS; field allow-list; personal data removed; API key kept in a server secret, never in code	SEC-11, LEG-3, AI-9
B5 Server to backup storage	Nightly backup job	Encrypted before it leaves the server; storage in the UAE; separate credentials with write-only access for the job	SAF-8, LEG-3

6.2 Access Matrix

Prepared by: Rashid Alzaabi

Table 6-2 shows who may perform each controlled action, for all nine roles, the Applicant and the AI services. The rule is default deny: anything not marked is refused, and the refusal is audited (FR-7.A1). "Own" means the user's own department, and the other conditions in the cells are checked by the system. DR, AC, AD, PT, MS, FO, SO, SA and UM are the nine roles of Table I-1, and "-" means refused. Sources: Phase 3 Table 5.1-1, SEC-5, FR-7.R1, FR-7.R3, FR-7.A3, FR-12.1, FR-12.9 and BR-2 to BR-11.

Table 6-2: Access matrix (default deny)

Action	DR	AC	AD	PT	MS	FO	SO	SA	UM	Applicant	AI services
Apply for an account, give or refuse consent	-	-	-	-	-	-	-	-	-	Yes	never
Activate accounts; assign roles and departments	-	-	-	-	-	-	-	Yes	-	-	never
View own personal data, request correction, withdraw consent	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-	never
Search the catalogue	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-	rank only
See financial values (SEC-5)	-	own assets	Yes	Yes	own work	Yes	totals only	-	Yes	-	never
Register, edit, publish or withdraw an asset	-	own	-	-	-	-	-	-	-	-	suggest only
Accept AI classification suggestions	-	own	-	-	-	-	-	-	-	-	never
Submit, edit or cancel a request	own	-	-	-	-	-	-	-	-	-	draft only
Choose a match for an open need	-	-	-	Yes	-	-	-	-	-	-	rank only
Release or decline a request	-	own assets, not own request	-	-	-	-	-	-	-	-	never
Approve or reject a released request	-	-	not own or released	second approval	-	-	-	-	-	-	never
Arrange a transfer and record handover	-	own assets	-	-	-	-	-	-	-	-	never
Confirm receipt or report a problem	own	own	-	-	-	-	-	-	-	-	never
Decide a transfer dispute	-	-	Yes	-	-	-	-	-	-	-	never
Report a defect	Yes	Yes	-	-	Yes	-	-	-	-	-	never
Request an inspection	-	Yes	-	-	Yes	-	-	-	-	-	never
Assign maintenance work	-	-	Yes	-	Yes	-	-	-	-	-	never
Record inspection and repair results	-	-	Yes	-	assigned	-	-	-	-	-	never
Propose an end-of-life action	-	own	-	-	-	-	Yes	-	-	-	recommend only
Safety review of an end-of-life proposal	-	-	-	-	-	-	not proposer	-	-	-	never
Approve an end-of-life action	-	-	not proposer	-	-	-	-	-	-	-	never
Maintain the approved vendor list	-	-	Yes	-	-	-	-	-	-	-	never
Maintain category values	-	-	-	-	-	prices, costs	weights, factors	-	-	-	never
Generate standard reports	own	own	Yes	Yes	-	Yes	Yes	-	Yes	-	never
Request an AI report	-	-	Yes	Yes	-	Yes	Yes	-	-	-	draft only
Approve an AI report	-	-	Yes	-	-	Yes	no financial figures	-	-	-	never
Use the assistant	Yes	-	-	-	-	-	-	-	-	-	answer, draft
Request a human review of an AI result	affected	affected	-	affected	affected	-	affected	-	-	-	never
Decide a human review	-	-	Yes	-	-	-	-	-	-	-	never
Read the audit log	-	-	Yes	-	-	-	-	Yes	-	-	never
Manage AI model versions and thresholds; set approval limits	-	-	-	-	-	-	-	Yes	-	-	never

Holding several roles does not let a person skip a separation rule: the checks compare user IDs across the stages of the same record, not role names (BR-4). An account that is suspended or has withdrawn consent is refused on its next call, even in an open session (FR-6.7, FR-6.10).

6.3 Audit Log

Prepared by: Rashid Alzaabi

The audit log records every action listed in FR-7.A1. Each entry is written in the same database transaction as the change it describes; if the entry cannot be written, the change is rolled back and the user sees an error (FR-7.A6). Table 6-3 lists the fields.

Table 6-3: Audit entry fields

Field	Content	Example
Who	Internal user ID and the role used (never a name or e-mail, SEC-8)	user 1047, Administrator
What	Action and record type and ID	APPROVE, request R-2026-0391
When	Date and time to the second, server clock (UAE time)	27/10/2026 14:02:31
Before	Old values of the changed fields	state: Released
After	New values of the changed fields	state: Approved; reserved: 2
Reason	Reason entered by the user, where required	Not needed for approval
AI link	AI output ID when the action used an AI result	AI-2026-08812
Chain	Hash of the previous entry and of this entry	9f3a...c21

- **Cannot be changed.** The application connects to the audit table with a database account that can only insert and read; no screen edits or deletes entries, even for System Administrators (FR-7.A2). Each entry stores a hash of the previous entry, so any change outside the application breaks the chain and is found by the nightly check.
- **No secrets or personal values.** Passwords, recovery secrets and personal-data field values are never stored; for those fields the entry records only that the field changed (FR-6.12, FR-7.A1).
- **Who can read it.** Only System Administrators and Administrators, searching by user, record, action and date (FR-7.A3). Every search of the log is itself audited.
- **How long.** No entry is deleted until university IT sets the retention period (FR-7.A5, TBD-4).

6.4 Data Storage and Encryption

Prepared by: Rashid Alzaabi

Table 6-4: Where data is stored and how it is protected

Data	Where	Protection
Business records, text vectors, job queue	MySQL 8 on an encrypted disk volume	Encryption at rest; private network; least-privilege database accounts (SEC-6)
Passwords	USER table	Stored only as salted PBKDF2 hashes, Django default [6] (SEC-6)
Photos and documents	File volume outside the web root	Image or PDF, 10 MB or less, type checked by content; served only after a permission check (SEC-13)
Personal data	USER and CONSENT_RECORD only	Five fields only (SEC-7); removed on consent withdrawal (SEC-8); never sent to AI services (SEC-11)

Data	Where	Protection
Secrets (database password, provider API key)	Server environment file, readable by the service account only	Never in the code repository; changed if exposed
Backups	Off-server storage in the UAE	Encrypted before upload (AES-256); separate key held by the System Administrator and one other named person

6.5 Backup and Recovery

Prepared by: Rashid Alzaabi

Table 6-5: Backup and restore plan

Item	Design
How often	Every night at 02:00 a full database dump and a copy of new or changed files; at most 24 hours of data can be lost (SAF-8)
Where	Encrypted copy on off-server storage in the UAE (LEG-3); the last 14 daily and 8 weekly copies are kept
Check	After each backup the job checks the file size and checksum and notifies the System Administrator if it failed (FR-7.N5)
Restore steps	1. Start a clean virtual machine from the deployment files. 2. Download and decrypt the last backup. 3. Load the database dump and the files. 4. Start the containers and run the health check. 5. Rebuild missing text vectors in the background (matching shows "not ranked" until done).
Restore test	Every week the System Administrator restores the latest backup into a separate test machine, times it against the 60-minute target (QUA-6), checks the record counts and the audit hash chain, and records the result
Estimated time	About 35 minutes: 10 to start the machine, 10 to download and decrypt, 10 to load, 5 to check

6.6 Integration Points

Prepared by: Rashid Alzaabi

Table 6-6: Integration points and security features

Integration or feature	Status	How it is secured or provided
AI Provider (language model)	Prototype	Through AIGateway only: HTTPS, field allow-list, secret API key, time limits, recorded outputs (DR-4)
Backup storage	Prototype	Encrypted upload, write-only job credentials, UAE location (Section 6.5)
E-mail copies of notifications	Later (TBD-6)	Desirable copy of in-application notifications; no change to the notification module interface
University single sign-on	Later (ORG-3, TBD-3)	Replaces the local sign-in after university IT approval; roles stay in ReSource
Sign-in, passwords, sessions (SEC-1 to SEC-4)	Prototype	Django authentication and session framework, lockout rule in the Accounts module, 30-minute idle expiry
Permissions and encryption (SEC-5, SEC-6)	Prototype	Access matrix checks in every service; HTTPS; encrypted volume and backups
Input handling (SEC-13)	Prototype	Django forms and parameterized queries; template escaping; upload type and size checks [12]

6.7 Draft Data Protection Impact Assessment

Prepared by: Rashid Alzaabi

LEG-2 requires a data protection impact assessment to be drafted during design, because AI matching evaluates requests automatically (UAE PDPL [13]). Table 6-7 is the draft. The university must approve it before any real personal data is used (LEG-2, ORG-2, TBD-11).

Table 6-7: Draft data protection impact assessment

Item	Assessment
Personal data processed	Name, university ID, university e-mail, department and role of each user (SEC-7); consent records. No sensitive personal data.
Purpose	Running accounts, permissions, approvals and the audit trail for campus asset exchange.
Automated evaluation	Matching ranks assets for a request; it does not evaluate people. Department name and size are not inputs (LEG-9); every match is decided by a person (H-2), and any affected user can ask for a human review (SEC-10).
Transfers outside the UAE	None of personal data: the AI Provider receives only allow-listed asset and request fields (SEC-11, LEG-3). Whether non-personal asset text may go to the provider is confirmed under TBD-9.
Main risks	1. Personal data typed into free text reaches the provider. 2. A user sees records of another department. 3. Audit history conflicts with an erasure request. 4. A data breach of the server or backups.
Measures	1. Gateway redaction and a warning on free-text fields. 2. Default-deny checks on every call and tests per role. 3. Audit entries hold only internal user IDs, so erasing the user record leaves no name (SEC-8). 4. Encryption at rest and in transit, encrypted backups, lockout and session limits.
Remaining risk and decision	Low after the measures, subject to university IT and legal review (TBD-4, TBD-11).

7. Mockups

7.5 Maintenance Tracking

Prepared by: Rashid Alzaabi

Maintenance Staff record the inspection of a Chemistry fume hood whose airflow sensor is faulty. The asset history shows the cumulative cost against the 50% review threshold (FR-5.10, FR-5.11). The ActionRecommender suggests repair with Medium confidence, gives its reason and main limitation, and never changes the asset by itself; the staff member chooses the next step.

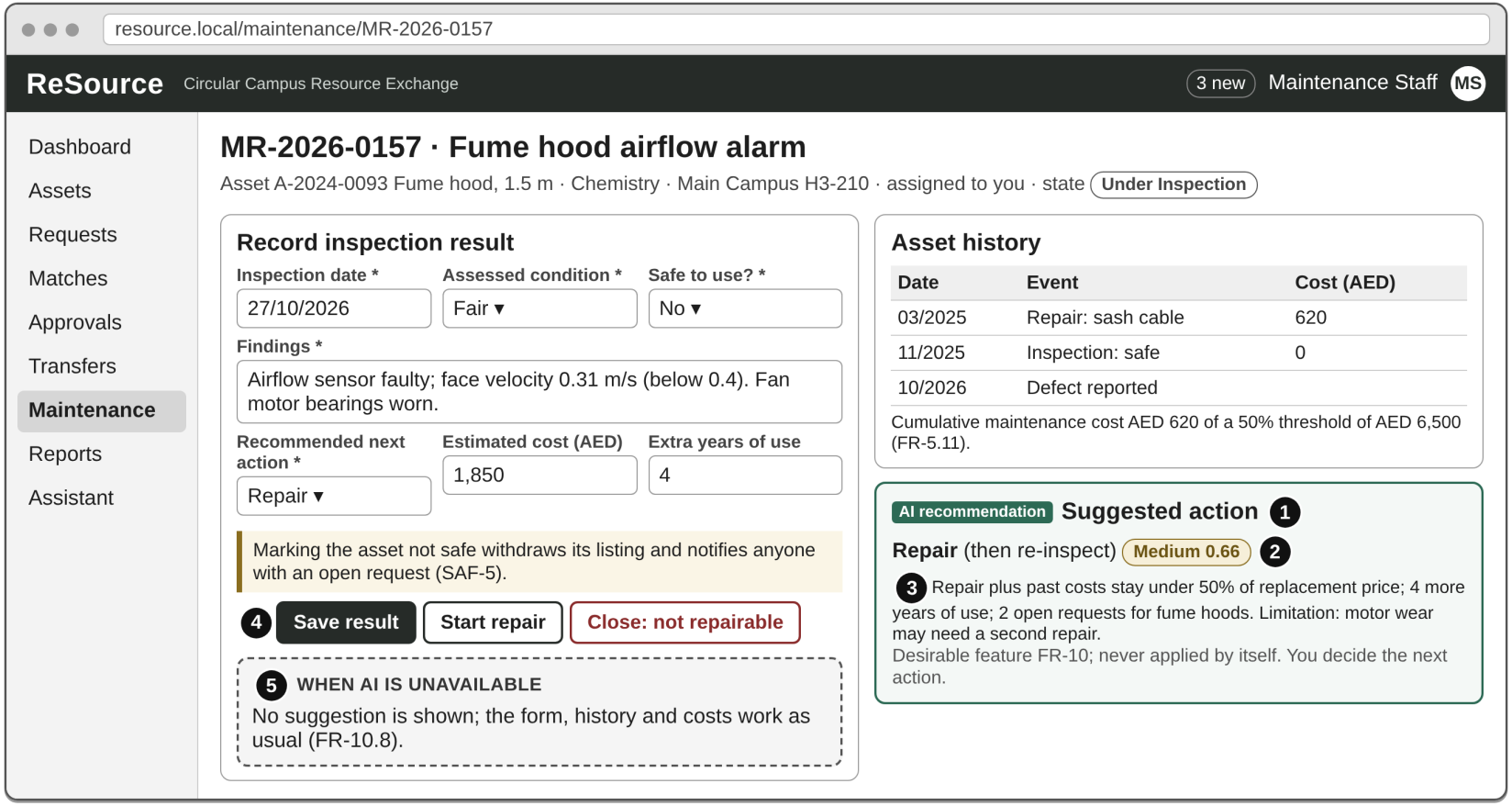


Figure 7-5: Maintenance Tracking screen

7.6 Sustainability Dashboard

Prepared by: Rashid Alzaabi

The Sustainability Officer sees the September indicators for the whole campus (the screen shows five of the nine), their trend, the share of default values and the "estimate" labels (FR-13.7, FR-13.12). AI highlights point out large changes and are calculated from the same indicator values.

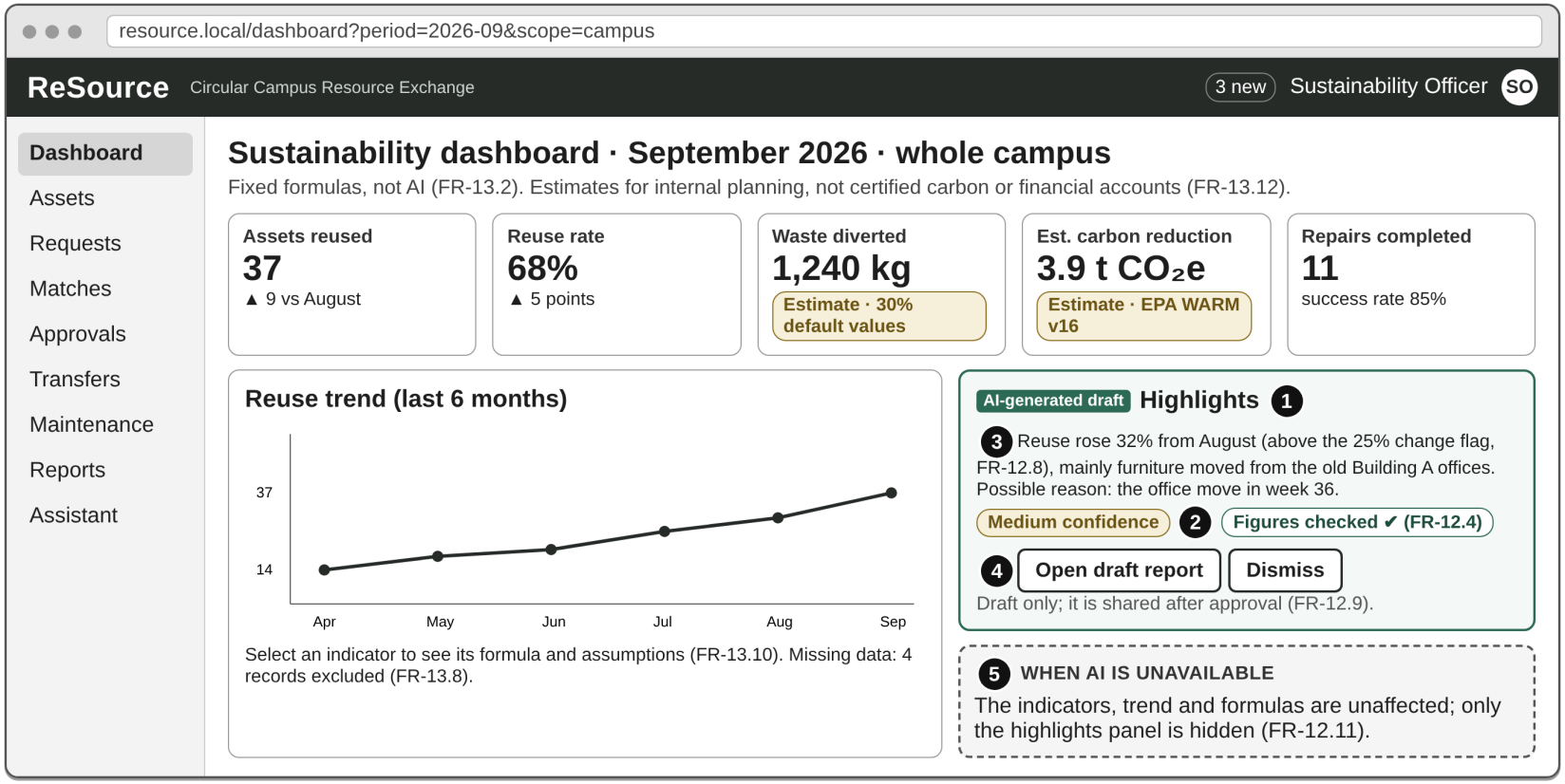


Figure 7-6: Sustainability Dashboard screen

7.7 AI Report Review

Prepared by: Rashid Alzaabi

The Finance Officer reviews the AI report draft for September. Every figure has been checked against the standard reports, the assumptions are listed, and the report stays a draft until it is approved (FR-12.9).

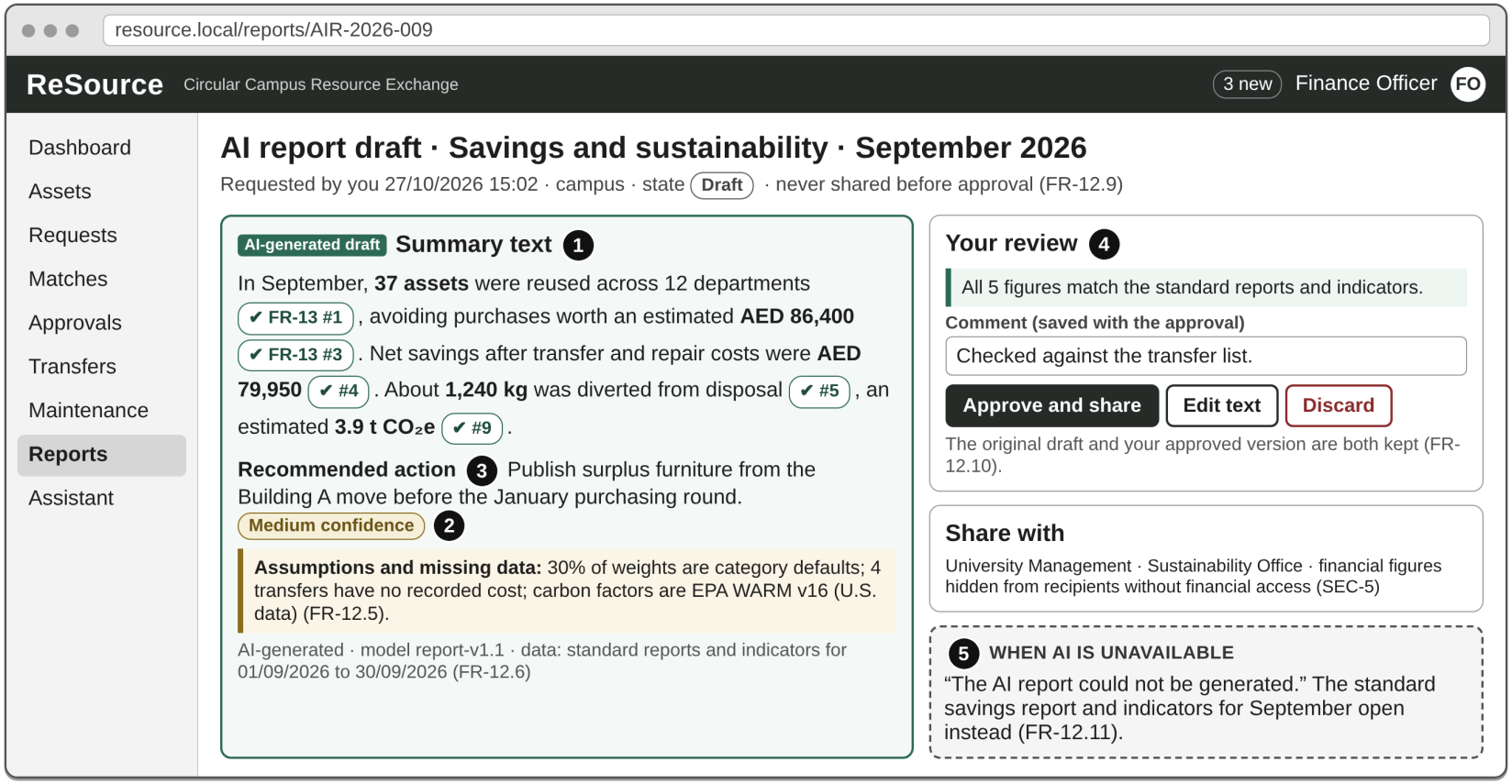


Figure 7-7: AI Report Review screen

9. Class Diagrams

9.4 Notifications, Audit and Sustainability

Prepared by: Rashid Alzaabi

A Notification is received by one User and points to the record it is about (FR-7.N1). An AuditEntry is performed by a User, or by the system when no user is linked, and records the AI decision when one was involved; `verifyChain` checks the hash chain of Section 6.3 (FR-7.A6). A Category has versioned CategoryValues, and each SustainabilityImpact uses the value versions it was calculated with (FR-13.5), for the whole campus or for one Department.

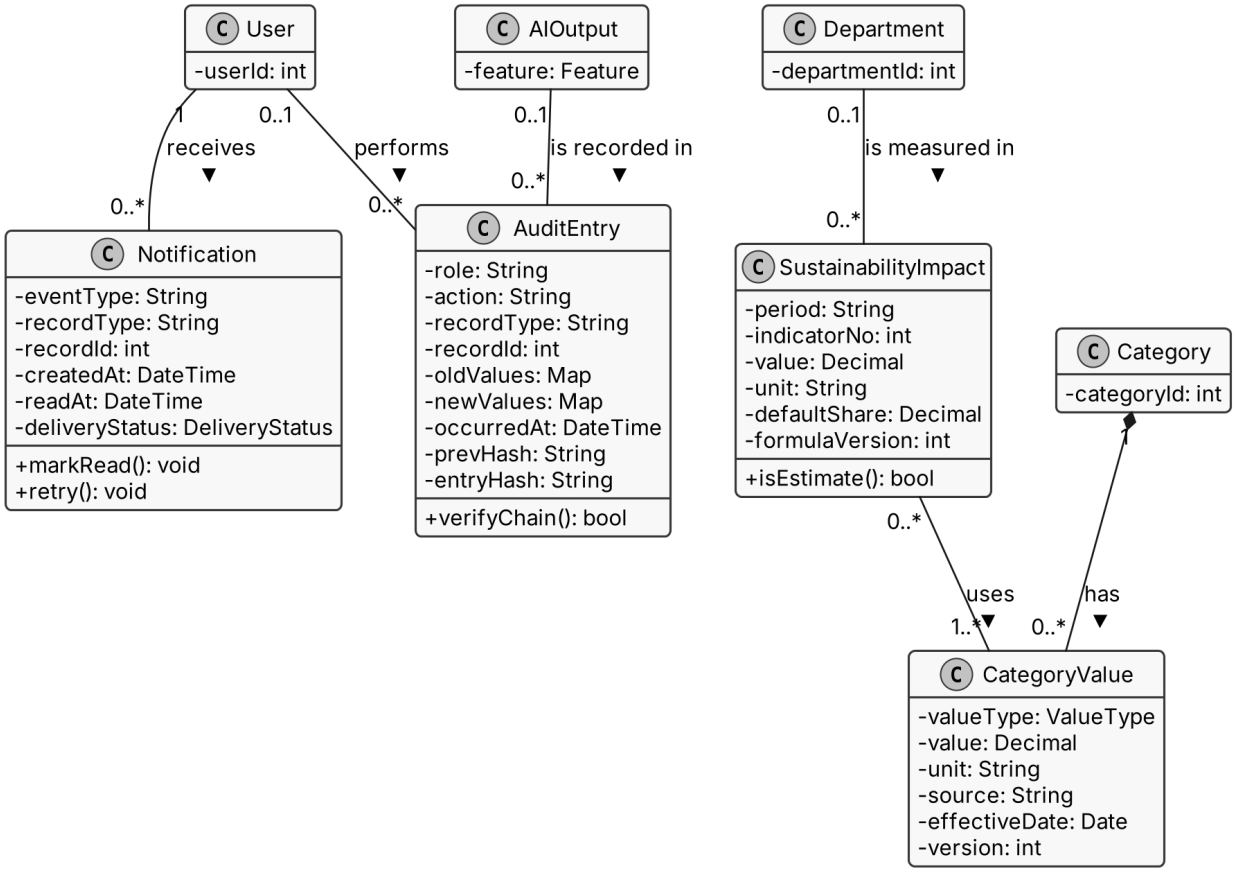


Figure 9-6: Class diagram, notifications, audit and sustainability

10. Sequence Diagrams

10.5 Generate an AI Report (UC-12)

Prepared by: Rashid Alzaabi

Figures are calculated with fixed formulas before the AI is called; the language model only writes the text, and ReportGenerator checks every number against the standard figures (FR-12.4). The report stays a draft until an authorized reviewer approves or discards it (FR-12.9).

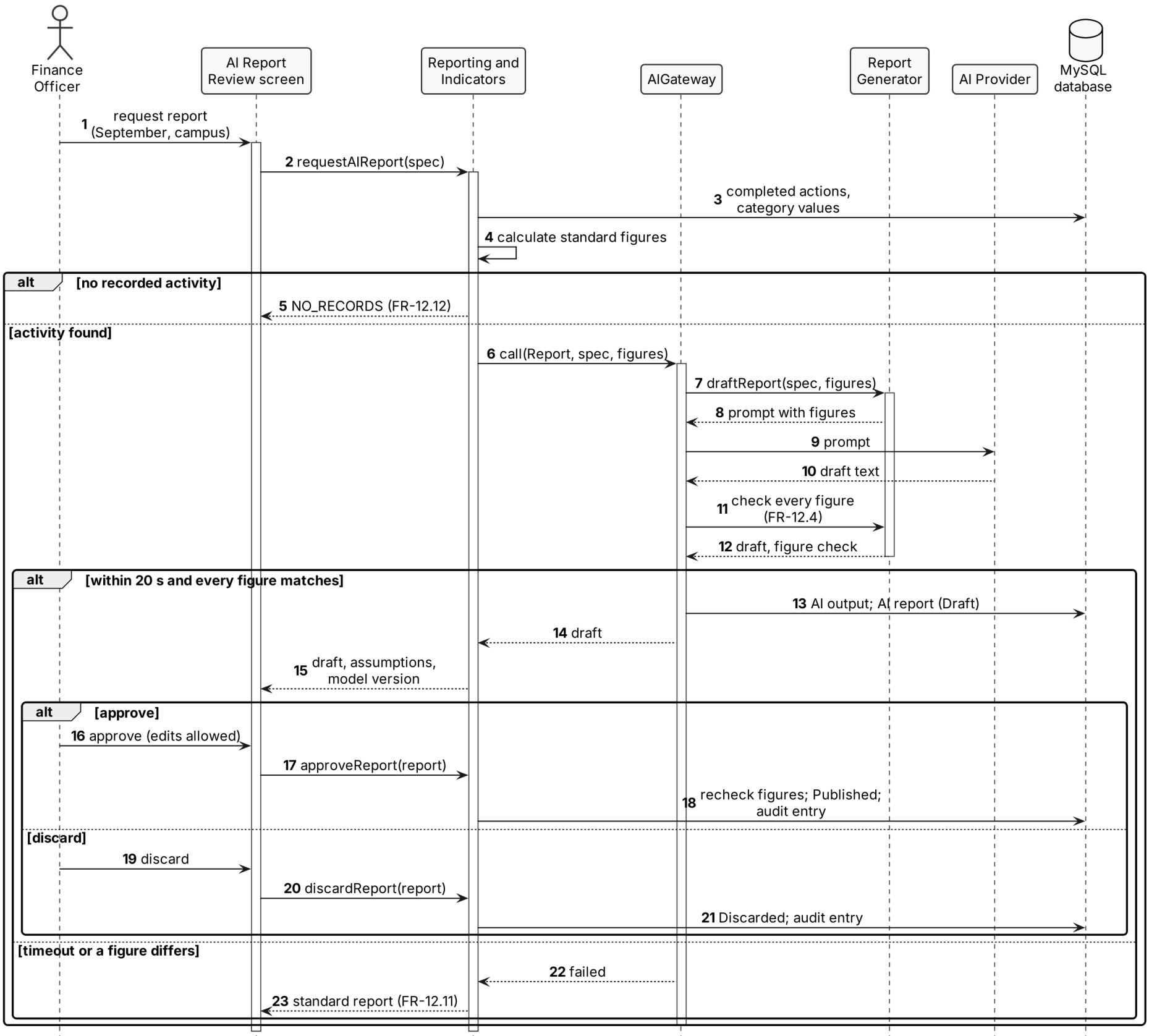


Figure 10-6: Generate an AI report

10.9 Record Maintenance or Repair (UC-6)

Prepared by: Rashid Alzaabi

A defect report goes to the Administrator, who assigns a Maintenance Staff member. While the work is open the asset is Under maintenance and hidden from search (FR-5.9). A repair restores the previous state with the new condition; a repair that brings the total cost to 50% of the replacement price, or a request closed as not repairable, flags the asset for end-of-life review and notifies the Asset Coordinator and the Sustainability Officer (FR-5.11, FR-5.12).

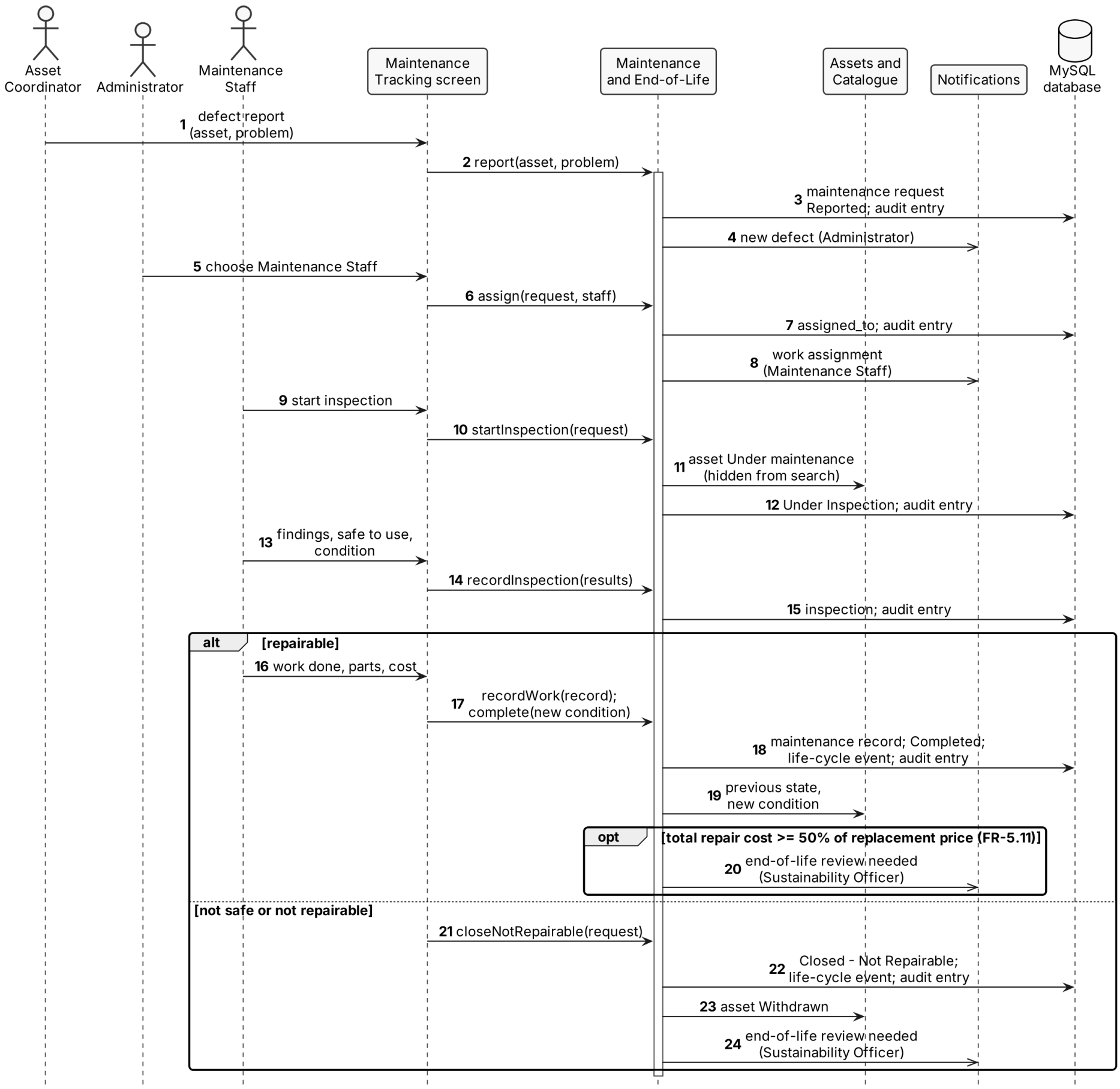


Figure 10-10: Record maintenance

10.10 Propose and Complete an End-of-Life Action (UC-7)

Prepared by: **Rashid Alzaabi**

The proposal is refused while the asset has a reservation, a transfer or open maintenance (FR-5.21). The AI advice is shown to the proposer but never decides. A second Sustainability Officer and then an Administrator must approve (BR-8); a rejection at either stage returns the asset to its previous state and nothing runs after it. The completion record needs an approved vendor and, for hazardous items, a certificate (FR-5.25, FR-5.30).

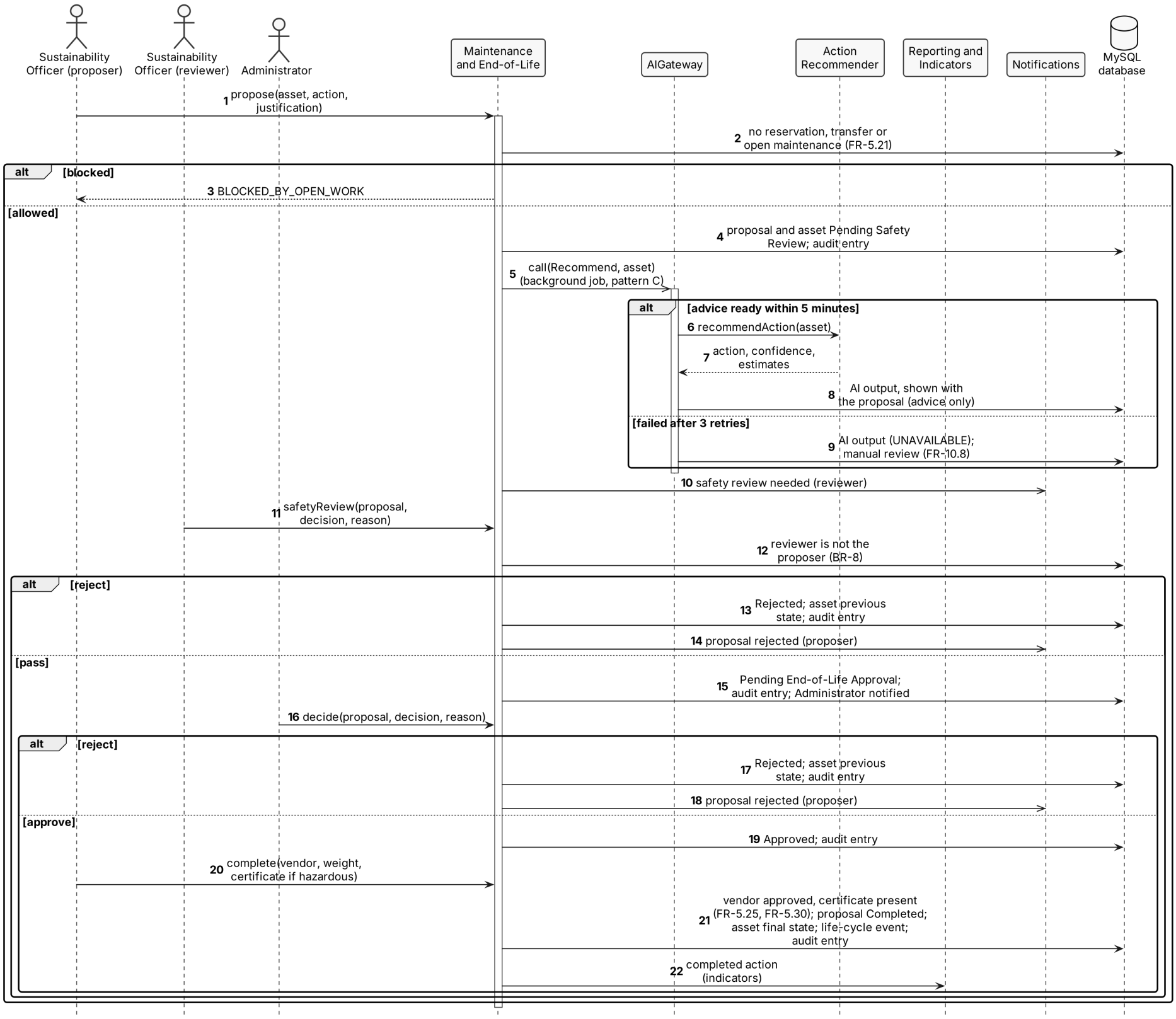


Figure 10-11: End-of-life action

11. Activity Diagrams

11.3 Maintenance and End of Life (UC-6, UC-7)

Prepared by: Rashid Alzaabi

A repair whose cumulative cost reaches 50% of the replacement price, or a request closed as not repairable, flags the asset for end-of-life review (FR-5.11, FR-5.12). The proposal passes a safety review by a second Sustainability Officer and an approval by an Administrator before the completion record is entered (BR-8). The AI advice is shown to the proposer but never decides.

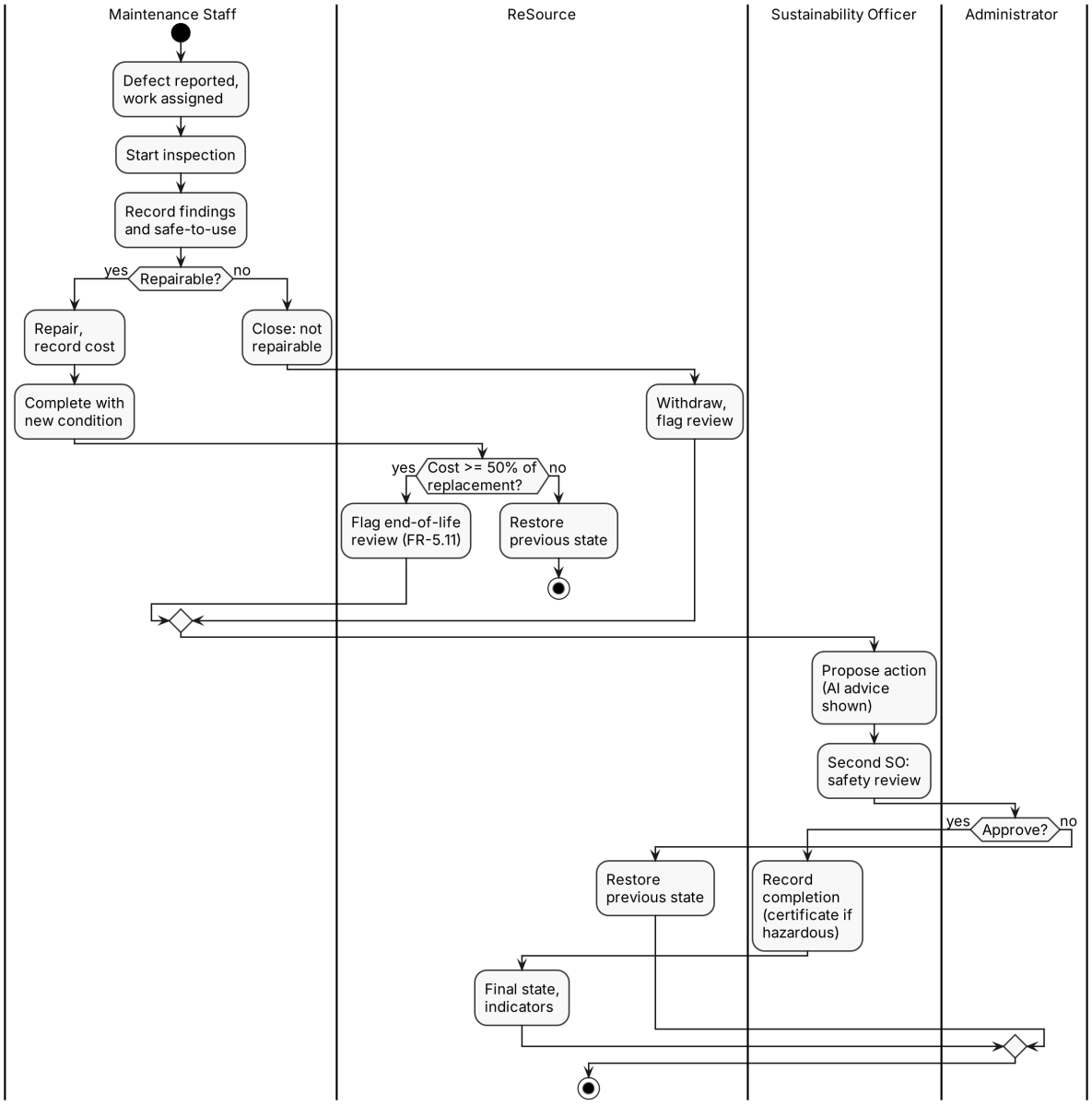


Figure 11-3: Maintenance and end of life

12. State Diagrams

12.2 Asset Maintenance and End of Life

Prepared by: Rashid Alzaabi

While an asset is Under maintenance, Pending Safety Review or Pending End-of-Life Approval it is hidden from search and matching and cannot be requested (FR-5.9, FR-5.27). A rejection returns it to the state it had before (FR-5.28).

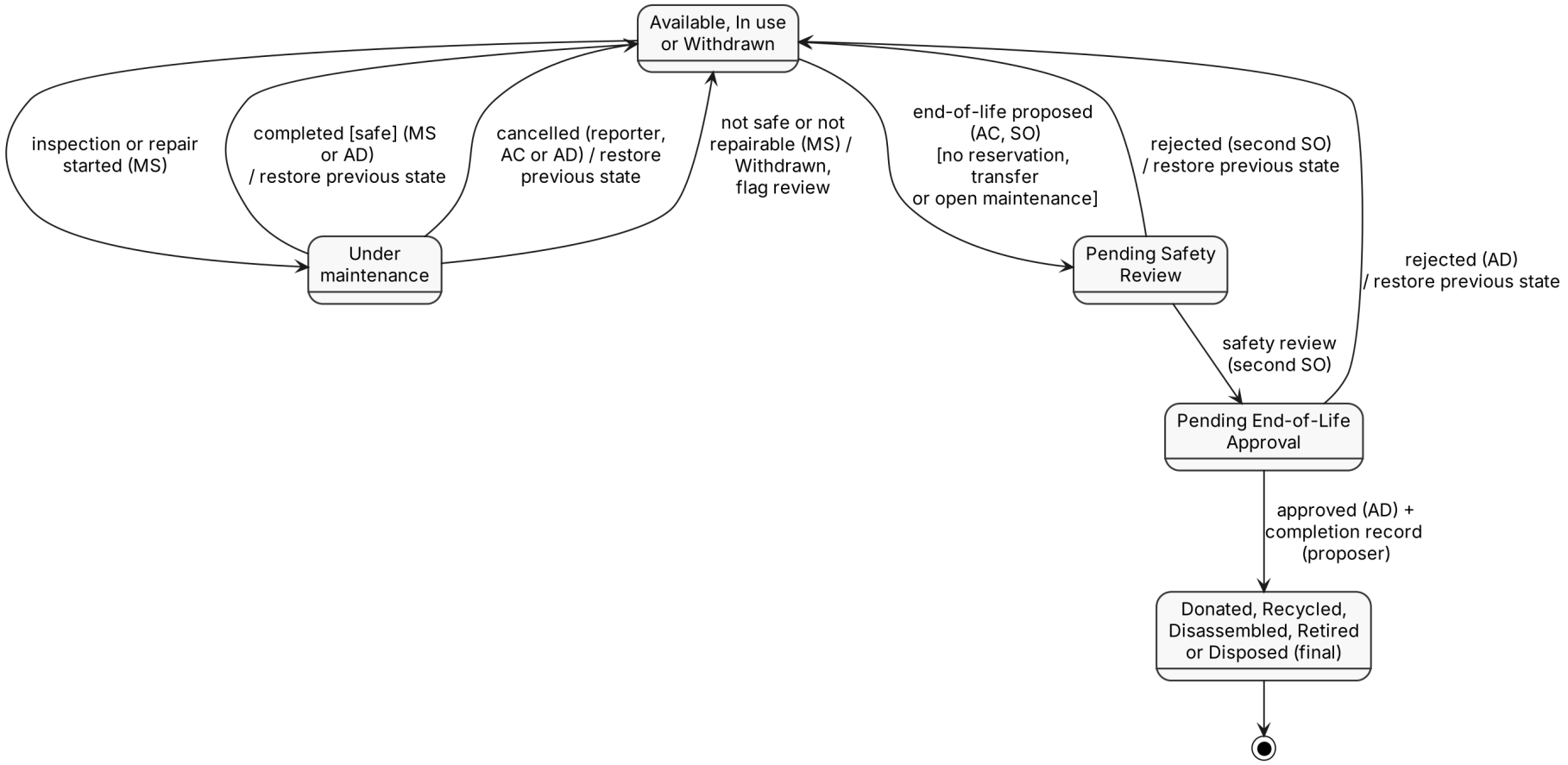


Figure 12-2: Asset maintenance and end-of-life states

12.5 Maintenance Request

Prepared by: Rashid Alzaabi

Only the assigned Maintenance Staff member or an Administrator can record inspection and repair results (FR-5.15). Completing requires the new condition (FR-5.8). Every cancel transition is made by the reporting user, the Asset Coordinator or an Administrator, with a reason (FR-5.13).

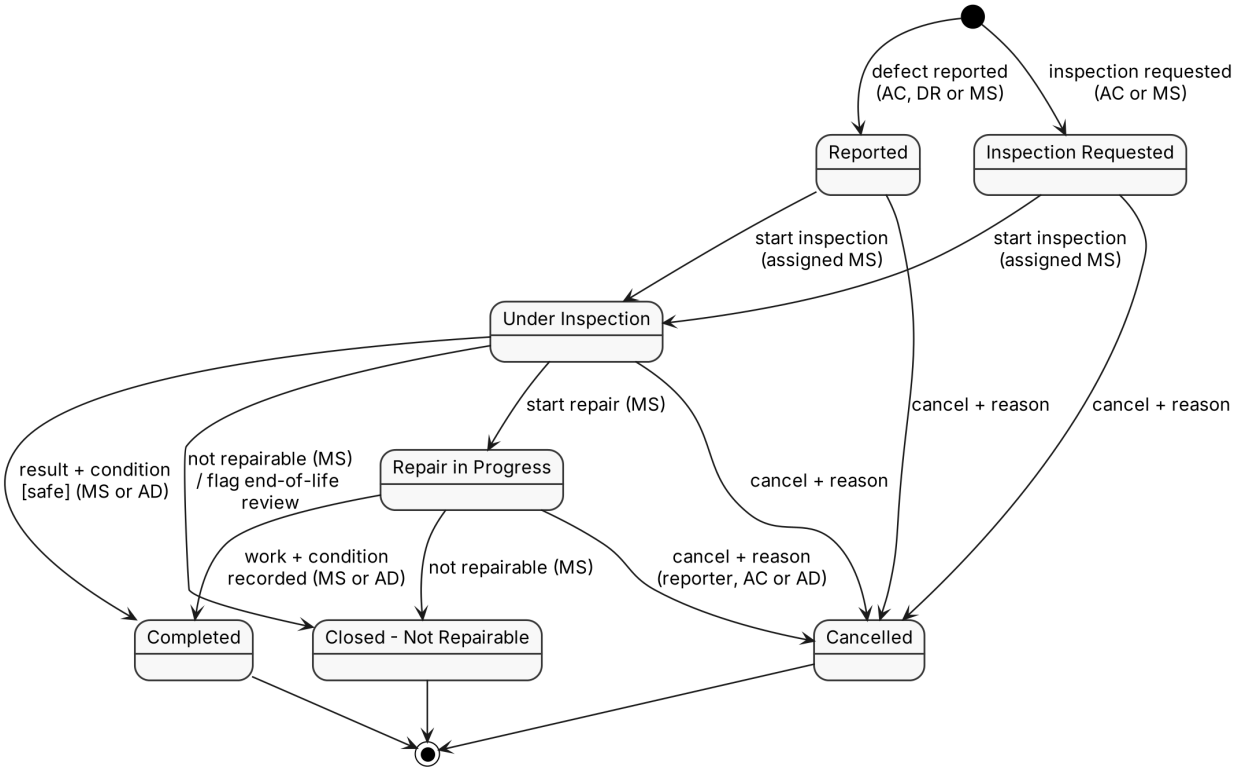


Figure 12-5: Maintenance request states

12.6 End-of-Life Proposal

Prepared by: Rashid Alzaabi

An end-of-life action passes two approval stages by two different people before the asset reaches its final state (BR-8, SAF-3). A rejection at either stage returns the asset to the state it had before (FR-5.28).

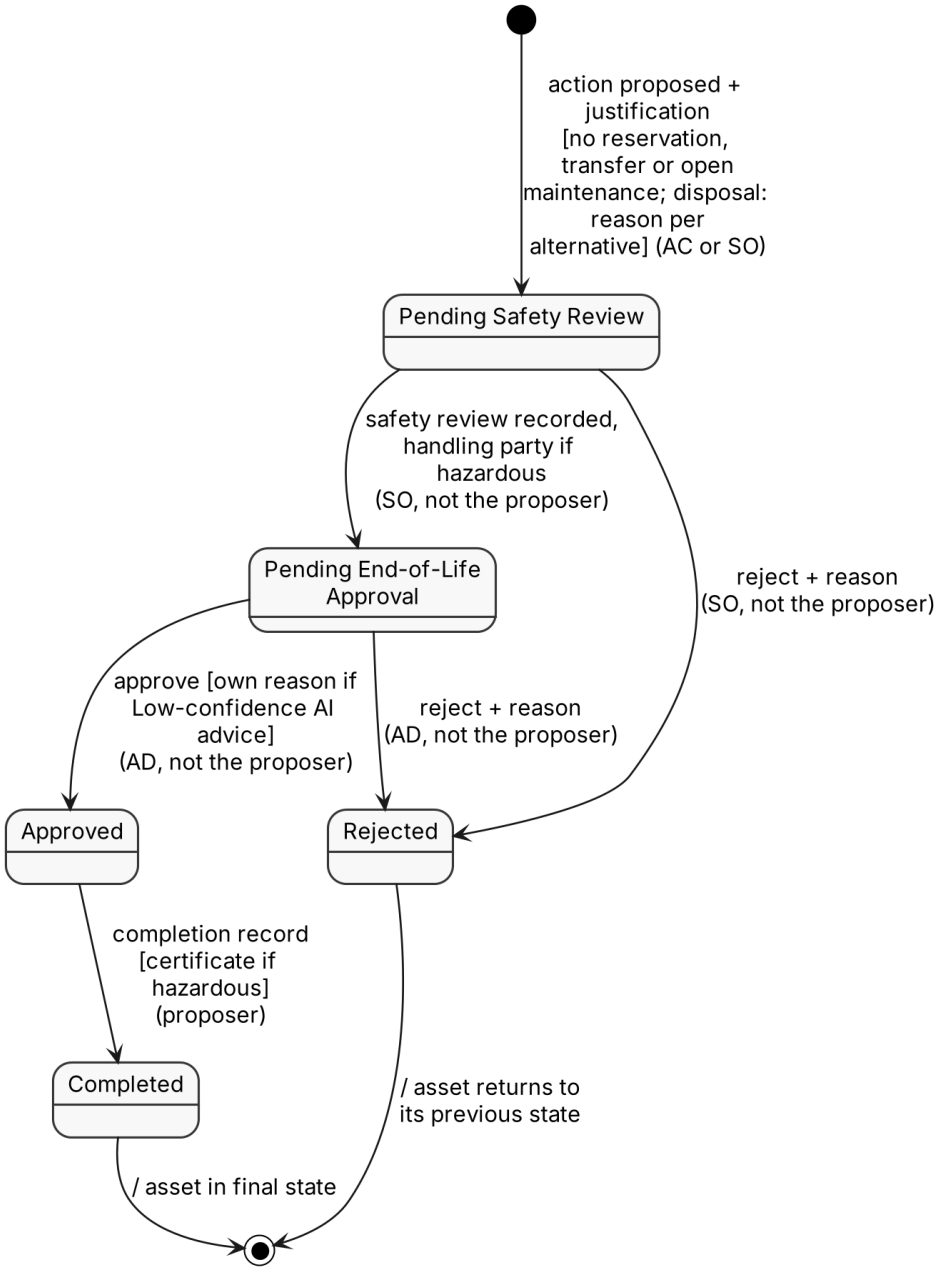


Figure 12-6: End-of-life proposal states

13. Component and Deployment Diagrams

13.2 Deployment Diagram

Prepared by: Rashid Alzaabi

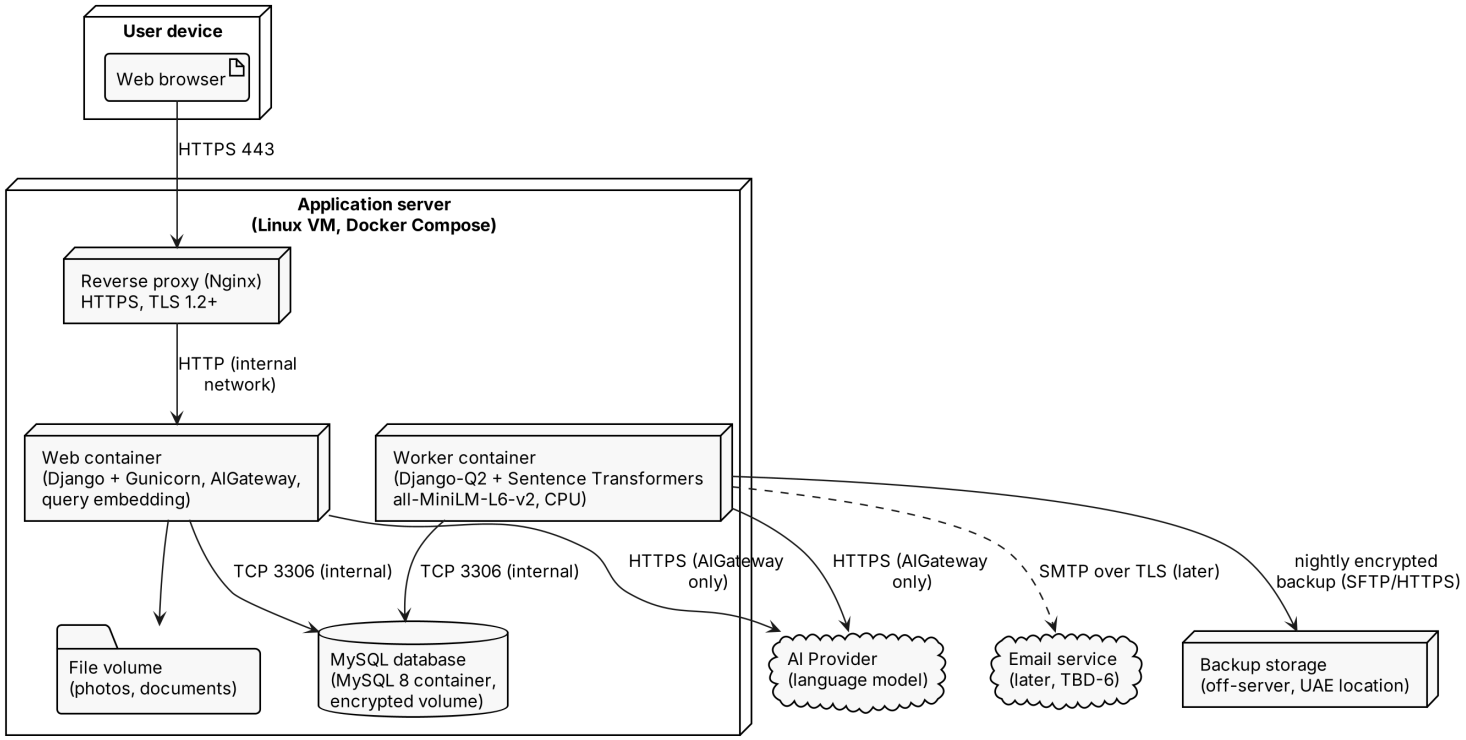


Figure 13-2: Deployment diagram

Table 13-1: Deployment nodes

Node	Runs	Notes
Reverse proxy	Nginx	Only public entry point, port 443; HTTP redirected; TLS certificate
Web container	Django, Gunicorn, AIGateway, query embedding	Serves pages and module services; restarts automatically
Worker container	Django-Q2, embedding model, scheduled jobs	Embeddings, retries, backups, monitoring
Database container	MySQL 8	Internal network only; encrypted volume
File volume	Photos and documents	Outside the web root; served after a permission check
Backup storage	Encrypted copies	UAE location; 14 daily and 8 weekly copies

18. Sustainability Calculations

Prepared by: Rashid Alzaabi

The nine indicators are calculated with the fixed formulas of Phase 3, Table 2.13-1, never with AI (FR-13.2). This section designs how the values are found, stored and reproduced.

18.1 Calculation Pipeline

Prepared by: Rashid Alzaabi

1. **Trigger.** A completed transfer, repair, refurbishment, donation, recycling, disassembly, retirement or disposal (FR-13.3), a correction record, or a nightly run.
2. **Find each value.** For each record and value (replacement price, handling cost, weight, additional years, emission factor): use the actual value on the record; otherwise the category default in force on the action date; otherwise list the record under "Missing data" and leave it out (FR-13.6, FR-13.8).
3. **Calculate.** Apply the formulas per indicator, counting each asset unit once per period. A rate with a zero denominator is shown as "not available", not as 0.
4. **Store.** Save one SUSTAINABILITY_IMPACT row per period, scope and indicator with the share of default values, and link the CATEGORY_VALUE versions used in IMPACT_VALUE_USED (FR-13.5).
5. **Label.** Mark the indicator "Estimate" when any record used a default, and show the share (FR-13.7).

18.2 Values and Sources

Prepared by: Rashid Alzaabi

Table 18-1: Category values

Value	Unit	Maintained by	Source
Replacement price	AED per unit	Finance Officer	Recorded price, else category default (FR-13.4)
Storage and handling cost avoided	AED per unit	Finance Officer	Category default (TBD-8)
Weight	kg per unit	Sustainability Officer	Recorded weight, else category default (FR-5.32)
Additional years of use	years	Sustainability Officer	Maintenance Staff estimate, else category default
Emission factor	kg CO2e per kg	Sustainability Officer	EPA WARM version 16 source-reduction factors [15], Phase 3 Table 2.13-2

Every value has a source, an effective date and a version; changing a value creates a new version, so a past result can always be reproduced with the versions it used (FR-13.5).

18.3 Worked Example

Prepared by: Rashid Alzaabi

Two oscilloscopes of A-2026-0412 are transferred to Physics and receipt is confirmed. The recorded values are a replacement price of AED 4,200 and a weight of 2.6 kg per unit; the category defaults are an avoided storage and handling cost of AED 18 per unit (the Phase 2 assumption [5]) and 4 additional years of use per unit; the transfer cost is AED 120; the emission factor for "Other electronic equipment" (WARM Mixed Electronics) is 22.92 kg CO2e per kg.

Table 18-2: Worked indicator example

#	Indicator	Calculation	Result
1	Assets reused	2 units	2
3	Avoided purchases	2 x AED 4,200	AED 8,400
4	Financial savings (net)	8,400 + 2 x 18 - 120 - 0 maintenance	AED 8,316 (estimate)
5	Waste diverted	2 x 2.6 kg	5.2 kg
8	Asset-life extension	2 x 4 years (default)	8 years (estimate)
9	Estimated carbon reduction	5.2 kg x 22.92 kg CO2e per kg	119.2 kg CO2e

Indicators 4 and 8 use defaults and are labeled "Estimate". The rates (indicators 2 and 6) and the repair figures (indicator 7) are calculated over the whole period, not per transfer. All figures are estimates for internal planning and are not certified carbon or financial accounts (FR-13.12).

AI Contributions

Rashid Alzaabi

Prepared by: Rashid Alzaabi

I used an AI assistant (Claude, Anthropic) to help draw the notification, audit and sustainability class diagram (Section 9.4), the sequence diagrams in Sections 10.5, 10.9 and 10.10, the maintenance activity and state diagrams (Sections 11.3, 12.2, 12.5 and 12.6), the deployment diagram (Section 13.2) and the screen mockups in Sections 7.5 to 7.7 from my design, and to check the permissions and sustainability figures against the Phase 2 and Phase 3 reports. I wrote the section content and reviewed every figure.

References

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[2] University, "Phase 4: Architecture Design," COSC 336 lab slides, University, Abu Dhabi, UAE, Fall 2026.

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